BKT-plasma2D 9.21: Session Summary

Correlation analysis of the 1210-point scan · Sep 29 – Oct 1, 2026

1. The 9.21 folder

Julia Monte Carlo of a 2D ±1 plasma (Ewald log Coulomb + hard core σ + "Majorana" attraction $-\log \cosh[(E0/T)e^{-r/\xi_F}]$, $\xi F = 10\sigma$), run on HTCondor over 10 ρ* × 11 T* × 11 E0* = 1210 states, N = 50, 1000 saved frames each. Raw coordinates in *Data/*, read with *read_scan.py*.

Files	Role
Setup.jl, Plasma_Interaction.jl, Monte_Carlo.jl	Initial state, Ewald + Majorana energies, Metropolis engine
Task_Animation_Dielectric.jl	Runs the scan and writes raw data (no animation or ε, despite the name)
run_scan_921.sh, scan_921.sub	Cluster jobs: one density (121 states) per job
one_over_epsilon0.jl	Standalone 1/ε formulas (not used by the scan)
read_scan.py, Analysis.ipynb	Python reader; original animation / correlation notebook

2. Conventions agreed

- **No 1/N:** $S_{NN}(k) = \langle|\Sigma\, e^{ik\cdot r}|^2\rangle$, $S_{QQ}(k) = \langle|\Sigma\, q\, e^{ik\cdot r}|^2\rangle$. Uniform gas → N, one compact droplet → N^2.
- **$1/\varepsilon = 1 - \kappa\, S_{QQ} / (2\pi\, m^2)$**, with L = 1, k = 2πm, κ = βg = 1/T*. Same as the PDF form; its N's cancel. BKT line: **$1/\varepsilon = 4T^*$**. Conductor: $S_{QQ}(k_{min}) = 2\pi T^*$.
- **$\rho^* = N\sigma^2/L^2$**, L = 1 always, so changing ρ* changes the particle size $a = \sigma = \sqrt{(\rho^*/N)}$. PDF screening: $(\kappa a)^2 = 2\pi\rho^*/T^*$. Bigger system = larger N ($L/\sigma = \sqrt{(N/\rho^*)}$).
- Everything below is post-processing of existing data; no cluster rerun was needed.

3. What was built (new files only; no original file modified)

File	Purpose
compute_structure_factors.py	Reads Data/ → S_{NN}, S_{QQ}, 1/ε for 10 shells, 5-block SEM, half-run drift test (~80 s)
Results/structure_factors.csv, S_kmin_series.npz	Outputs read by both notebooks
S_NN_Liquid_Gas.ipynb	**Task 2**: $S_{NN}(k_{min})$ vs T*, heatmaps, S(k) vs kσ (incl. new 4b/4c), MC-time series, tables
Dielectric_Phase_Diagram.ipynb → Results/bkt_crossings.csv	**Task 3**: 1/ε vs T*, T_{BKT} crossings, 9.6 check, $\beta g\varepsilon^{-1}$ map, combined phase diagram

Flags: *clustered* = $S_{NN}(k_{min}) > 5N$ (E0* = 0 pair gas sits at ~1.5–3N); *drifting* = first vs second half differ by > 3σ (179 of 1210 states).

4. Results: Task 2 (liquid–gas from S_{NN})

- **ρ* ≥ 0.02: clean clustering boundary.** $S_{NN}(k_{min})$ jumps from ~2N to 0.5–0.95 N^2. Onset lies along roughly constant **E0*/T* ≈ 0.6–0.7** (ρ* = 0.02–0.05), ≈ 0.75 (0.1), ≈ 1.05 (0.2).
- **Droplets are a dense liquid:** curves for ρ* = 0.01–0.2 collapse on one curve in kσ (T* = 0.25, E0* = 0.2). Guinier fit R ≈ 4.8σ and form-factor minimum (kσ ≈ 0.85 → R ≈ 4.5σ) agree; inside density ≈ $0.69/\sigma^2$, ~14× the box average at ρ* = 0.05.
- **ρ* ≤ 0.01: not equilibrated.** Acceptance 1–3% at low T*; the fixed step 0.05 is ~11σ at ρ* = 0.001, so moves break bound pairs. Clustering even rises with T* there (kinetic arrest).
- Correction: the droplet tail is a form factor, not a Porod k^{-3} law (droplet too small at N = 50).

5. Results: Task 3 (BKT line from 1/ε)

ρ*	T_{BKT} 9.6	T_{BKT} 9.21 (±SEM range)	Difference
0.001	0.2387	0.2400 (0.2395–0.2406)	+0.0013
0.005	0.2145	0.2152 (0.2141–0.2166)	+0.0007
0.01	0.1939	0.1897 (0.1874–0.1932)	−0.0042

- E0* = 0 reproduces 9.6 (differences ≪ grid step 0.02). T_{BKT} falls 0.240 → 0.161 over ρ* = 0.001–0.02, below 0.15 for ρ* ≥ 0.03. Dilute limit → 1/4 for all E0*.
- 89 crossings + 21 "conductor at all T" (NaN). Only **15 reliable**; crossing finder uses the last +→− change (3 single-state 1/ε spikes, down to −2.1).
- **E0* ≥ 0.25 is a droplet artifact:** a compact neutral droplet keeps 1/ε ≈ 0.8–0.95, so 1/ε = 4T* just lands near (1/ε)/4 ≈ 0.20–0.24.
- BKT = dielectric molecular gas (bound pairs, $\beta g\varepsilon^{-1} \geq 4$) ↔ conducting plasma (free charges, 1/ε → 0). Rough tricritical region where it meets the clustering boundary: E0* ≈ 0.1–0.2, ρ* ≈ 0.005–0.01, T* ≈ 0.19–0.22.

6. Comparison with the vortex_diagnostics.pdf schematic

Schematic curve	Your variables	Data
Molecular gas (βg = 6)	T* = 1/6, ρ* ≈ 0.01	$1/\varepsilon(k_{min})$ = 0.82 at (0.01, 0.17) vs ~0.80 ✓
Plasma (κa ≈ 1)	ρ*/T* ≈ 0.16, e.g. (0.05, 0.3)	1/ε = 0.09–0.12 vs Debye–Hückel 0.035 (weaker screening)
Liquid (κa ≈ 3.4)	ρ* ≈ 0.4–0.6	Not in grid; only inside droplets

Current shells reach only kσ ≈ 0.03–1.6; the schematic features (charge alternation ka ≈ 4, overscreening, packing peak ka ≈ 2π) need kσ up to ~10.

7. Open items and next steps

- **Reminder:** VS Code still shows the old S_NN_Liquid_Gas.ipynb without sections 4b/4c → Cmd+Shift+P → "Revert File" (do not save the stale tab).
- **Reminder:** fix the section 4 text in S_NN_Liquid_Gas.ipynb ("Porod ~k^{-3}" → droplet form factor).
- **Proposed, awaiting approval (Step A):** compute S_{QQ}, 1/ε, S_{NN} out to kσ ≈ 10 from existing data for chosen states, overlay the PDF model curves.
- **If rerunning (Step B):** step size ∝ σ (e.g. 0.5σ); add --N for N = 100/200 (size scaling, coexistence ∝ N^2); dense uniform-liquid runs at ρ* ≈ 0.4–0.6 with N ≈ 200–400.
- Further analysis: local-density histogram (binodal), Ornstein–Zernike fit (correlation length), Guinier fits (liquid density).